



Khaled Abdelhadi

Mechanical Engineer (30)

A seasoned simulation engineer with expertise in 1D-CFD and thermal management for automotive and electric mobility applications.

Open to roles in simulation, engineering, product development, management, design, technical leadership, and consulting.

Nationality: GERMAN

Languages: Arabic, English, German

Nuremberg, Germany

+49 17 666 955 398

kabdelhadi795@gmail.com

[linkedin.com/in/khaled-de](https://www.linkedin.com/in/khaled-de)

Mechanical Engineering
B.Sc.

Alfaisal University (SAUDI)

Design & Development in
Mechanical & Automotive Engineering
M.Eng.

Esslingen University (GERMANY)

Design of Industrial Piping Systems
Specialization

Larson & Toubro EduTech (ONLINE)

—

GT Suite - Creo - Spaceclaim - Revit -
MATLAB/Simulink - Python - MS Office -
JIRA - AGILE - ASPICE - CAD - CAM -
FEA - CFD - P&ID - ASME B31 / B36 -
BIM - PMP

TRATON R&D Germany GmbH / MAN Truck & Bus SE (Nuremberg, Germany)

1D Simulation Engineer / Expert in Vehicle Cooling Systems

May 2022 – February 2026 (3 yrs, 9 mos)

- Led thermal simulations for electric and diesel truck cooling systems using GT
- Developed and calibrated 1D cooling system models from 3D pipe geometry, achieving strong validation with 3D CFD results and physical test measurements.
- Optimized pump sizing, case studies and operating points for interconnected cooling circuits (battery, HV components, cabin, refrigerant), contributing to improvements in energy efficiency.
- Created configurable variant sub-models for different vehicle configurations.
- Documented simulation methodology, enabling faster onboarding and consistent model quality across the simulation team.

1D/2D Simulation Engineer in E-Motor Cooling

December 2021 – May 2022 (6 mos)

- Simulated e-motor electromagnetic behavior using ANSYS Maxwell and calibrated GT thermal models with AUDI Toolbox loss maps, achieving validated temperature predictions within sensor measurement tolerances.
- Identified critical hotspots and demagnetization risks from insufficient cooling.
- Discovered uncovered system losses, expanding the accuracy of the thermal model.

1D Simulation Student in E-Motor Cooling

October 2020 – March 2021 (6 mos)

- Built a bus e-motor 1D thermal model in GT-SUITE from scratch, including oil circuit design with conduction and convection connections, establishing the foundation used in all subsequent e-motor cooling studies.
- Created a detailed thermal model from 3D-to-1D conversion and validated transient thermal simulations against drive cycle test data.

Esslingen Rennstall (Esslingen am Neckar, Germany)

CFRP Monocoque Structure Simulation Member

March 2020 – August 2020 (6 mos)

- Optimized CFRP monocoque design for the Stallardo 19 vehicle using ALTAIR HyperWorks, targeting weight reduction while increasing torsional stiffness to meet structural requirements.
- Delivered production-ready BOM and 2D schematics for the Stallardo 20 chassis, translating simulation results into manufacturing documentation.

A123 Systems GmbH (Stuttgart, Germany)

Engineering Development & Management Student

January 2020 – August 2020 (8 mos)

- Developed project KPIs based on JIRA tickets within V-Cycle (ASPICE) phases, enabling data-driven milestone forecasting and improving team velocity.
- Analyzed BMS hardware/software architecture and mechanical battery design, gaining insight into automotive battery systems and safety testing protocols.

Alfanar Electric Company (Riyadh, Saudi Arabia)

Plastic Mold Design & Manufacturing Student

July 2017 – January 2018 (7 mos)

- Designed and manufactured steel molds using PTC Creo, programmed CNC, and performed plastic flow simulations to identify and resolve common molding defects.

Shell Eco Marathon Asia (Singapore)

CFRP Design & Manufacturing Student

June 2016 – June 2018 (2 yrs)

- Designed vehicle exterior in Creo, developed carbon fiber molds with CNC-machined Styrofoam, and led the composite layup and curing process, delivering a monocoque structure validated through tensile testing.